

System unit, motherboard, ports and expansion slots

One college period · 40–45 minutes · Inside System Unit

What you will learn

- Describe the system unit / CPU casing and its main parts
- Label motherboard components: CPU socket, RAM slots, BIOS, CMOS battery, buses, ports, expansion slots
- List serial, parallel, USB and HDMI ports
- State PCI / expansion slot purpose

System unit. The case that houses the motherboard, power supply, drives, cooling fans and ports — the “CPU box” of a desktop.

- Power supply (SMPS): converts 220 V AC to DC for components.
- Motherboard / main board: printed circuit that connects everything.
- Hard drive / combo drive bays.
- Ports on the back/front panel.
- Cooling: heat sink + fan, sometimes liquid cooling.

Motherboard components

Part	Function
Microprocessor socket	Holds the CPU
RAM slots	DIMM / SIMM modules of main memory
BIOS chip (ROM)	Starts the computer, POST, finds boot device
CMOS battery	Keeps date, time and BIOS settings when PC is off
Chipset	Northbridge/southbridge traffic between CPU, memory, I/O
Jumpers	Tiny connectors that set hardware options
Expansion slots	PCI / PCIe cards: graphics, Wi-Fi, sound
System bus	Electrical pathways for address, data, control

Ports

Port	Use
Serial (RS-232, old)	One bit at a time; mouse/modem in old PCs
Parallel (LPT)	Several bits at a time; old printers
USB	Universal, hot-pluggable: keyboard, disk, phone
HDMI	High-Definition Multimedia Interface — video+audio to TV/monitor
VGA	Analog video (older monitors)

REMEMBER IT

CMOS battery dead $\Rightarrow$ date/time reset every boot, sometimes "disk boot failure" until BIOS is set again.

HOW BIEK ASKS THIS

Short: components of the system unit. Long: motherboard with a sketch of CPU, RAM, slots, ports.

اردو میں سمجھیں

سسٹم یونٹ ڈیسک ٹاپ کا باکس ہے۔ مدر بورڈ سب کو جوڑتی ہے۔ پاور سپلائی بجلی دیتی ہے، پورٹس کی بورڈ اور مانیٹر لگاتے ہیں، USB عام پورٹ ہے۔ CMOS بیٹری تاریخ/وقت رکھتی ہے۔

CHECK YOURSELF

1. Which port carries video and audio together?
2. What does BIOS do at startup?

Answers: 1. HDMI · 2. POST and locate a bootable disk / OS

MCQs

1. The electrical path that carries data from one device to another is called:
 - a. Bus
 - b. CPU
 - c. Circuit only
 - d. Chip
2. USB stands for:
 - a. Universal Serial Bus
 - b. United System Board
 - c. Ultra Speed Bit
 - d. Uniform Serial Box

Key: 1=a, 2=a

Short questions (3 marks each)

1. Write the main components of a system unit.
2. What is CMOS battery used for?
3. Differentiate serial and USB ports.

Model short answers

1. Power supply, motherboard, CPU, RAM, storage drives, cooling, ports, expansion cards.
2. It powers CMOS memory so BIOS settings, date and time survive when the PC is unplugged.
3. Serial (old) sends one bit at a time; USB is a modern serial bus that is fast, hot-pluggable and powers devices.

Long questions (board Section C)

1. Describe the motherboard. Explain its components with a labelled diagram.